

Appendix A. Supplementary Material

This file contains the following tables:

1. Table A.1 - this table provides a summary of the types of anomalies in publicly available datasets. [M Ali: Remove all columns except "Dataset" and "Types of Anomalies"](#)
2. Table A.2 - this table provides a summary of the state-of-the-art studies based on RL.
3. Table A.3 - this table provides a summary of the state-of-the-art studies based on Gen AI.

Table A.1: Publicly available datasets for anomaly detection.

Dataset	Year	Format	F ¹	E ²	Domain	Types of anomalies	Setup
UNSW-NB15	2016	CSV, PCAP	49	~2,500,000	Network IDS	9 cat.: Fuzzers, Analysis, Backdoor, DoS, Exploits, Generic, Reconnaissance, Shell-code, Worms	Relatively balanced
SWaT	2016	CSV	51	~500,000	ICS (water treatment)	Normal and abnormal	Network traffic + data from 51 sensors/actuators
MIMIC-III	2018	CSV, SQL	47	61,532	Healthcare	Policy violation	Beth Israel Deaconess Medical Center
CIC-IDS2017	2018	CSV, PCAP	85	~3,000,000	Network	15 cat.: DoS, PortScan, Web (XSS, SQLi), Heartbleed, etc.	Generated via realistic simulation
N-BaIoT	2018	CSV	116	7,062,606	IoT botnet	2 cat.: Mirai, BASHLITE	9 commercial IoT devices
Bot-IoT	2019	CSV, PCAP	47	~3,000,000	IoT	4 cat.: DDoS, DoS, Reconnaissance, Information Theft	Realistic IoT env.; five scenarios via Argua and VMs (Node-RED)
ECU-IoHT	2020	CSV	8	111,207	Healthcare	4 cat.: ARP spoofing, DoS, Smurf, PortScan	IoHT monitoring devices
ToN-IoT	2021	CSV, PCAP, Logs	44	22,000,000	IoT/IIoT	9 cat.: DoS, DDoS, Backdoor, Injection, Password, Ransomware, XSS, MITM	Heterogeneous sources; large-scale network
IoT-23	2021	PCAP	23	~688,000	IoT malware/botnet	12 cat.: Okiru, Mirai, Torii, C&C-HeartBeat	Real IoT device traffic
CIC IoT 2023	2024	CSV, PCAP	47	~46,000,000	IoT	7 cat.: DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, Mirai	Multiple real IoT devices
CIC IoT DIAD 2024	2024	CSV (flow-based)	121	~3,600,000	IoT	7 cat.: DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, Mirai	105 devices in realistic IoT topology
CIC IoMT 2024	2024	CSV	39	440,315	IoMT	DDoS, DoS, Recon, MQTT, Spoofing	IoMT testbed (25 real + 15 simulated devices)

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Table A.1: Publicly available datasets for anomaly detection.

Dataset	Year	Format	F ¹	E ²	Domain	Types of anomalies	Setup
DataSense: CIC IIoT 2025	2025	PCAP, Logs (Elastic- search), CSV	~80	1,628,452	IIoT (Net- work + Sensor Logs)	7 cat.: Recon, DoS, DDoS, Web, MITM, Brute Force, Malware (Mirai) + Benign; 49 attack types	Time-synchronized network (PCAP) + sensor logs; Binary, Multiclass-8, and Multiclass-50 label setups; 40 devices

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¹ Number of features
² Number of examples.

Table A.2: Summary of state-of-the-art studies based on RL.

Ref.	Year	Dataset	RL algorithm	Task / objective	Results	Strengths	Limitations
[38]	2023	ToN-IoT (GPS / Modbus / Weather)	DQN	Red teaming / at- tack planning	Accuracy = 87.8% (Weather split)	Fast convergence; strong on Weather	Lower GPS accuracy (79.7%); Modbus weaker
[11]	2024	Smart grid simulator (custom)	DDQN (CNN & GRU)	Smart grid anomaly detec- tion	Accuracy = 95.3%	Accurate minority attack de- tection; real-time adaptabil- ity	Simulation only
[39]	2023	NSL-KDD	Deep SARSA	Network intru- sion detection	F1 = 84.6%	Superior minority-class detec- tion	Evaluated on older datasets
[40]	2023	Not specified	Actor-Critic	Adaptive threat detection / re- sponse	Conceptual	Supports adversarial learning; real-time policy updates	No dataset or empirical vali- dation
[41]	2023	IPMSRL (maritime OT simulation)	MAPPO & IPPO	Industrial / OT defense	Success rate = 99.5% (MAPPO)	Collaborative defense; config- urable environment	Domain-specific; complex simulator setup
[42]	2023	IoT-23	Recurrent RL (GRU & DQN)	IoT device threat detection	F1 = 80.6%	Captures sequential IoT dy- namics	High complexity; training overhead
[47]	2023	Traffic man- agement (simulated)	DRL (PPO / DDPG)	IoT-based traffic & cyber threat re- sponse	Reward +38% over baseline	Multi-agent coordination; real-time safety decisions	Environment-specific; no open dataset
[48]	2023	Enterprise network (sim- ulated)	DQN	Real-time net- work threat detection	Accuracy = 93.1%	Adaptive learning; dynamic policy updates	Limited generalizability to public benchmarks
[47]	2023	Smart city traffic (simu- lated)	DRL (policy gradient)	Smart city cy- ber-physical threat detection	Accuracy = 91.4%	Coordinated traffic-cyber re- sponse	Complex simulation; replica- tion difficult

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Table A.2: Summary of state-of-the-art studies based on RL.

Ref.	Year	Dataset	RL algo- rithm	Task / objec- tive	Results	Strengths	Limitations
[49]	2023	CyberBattleSim	DQL (reward shaping)	Red team attack simulation & de- fense	Steps-to-goal: 27% faster convergence	Reward-optimized adaptive planning	Limited generalization be- yond emulation
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Table A.3: Summary of state-of-the-art studies on Gen AI for cybersecurity.

Ref.	Gen AI alg.	Use case / domain	Dataset	Contribution	Limitations / challenges
[9]	GANs	Intrusion detection systems	CICIDS-2017; NSL-KDD; DARPA; CTU-13; RockYou	Taxonomy of GAN-based IDS approaches; highlights cGAN, DCGAN, WGAN for realistic intrusion patterns	Relies on legacy datasets (e.g., NSL-KDD); no real-world deployment or adversarial testing
[66]	GANs	Synthetic threat data generation	Simulated datasets	Uses GANs to generate synthetic threats for IDS; supports adversarial scenario emulation for training	Not validated on high-variance attack sets; no operational deployment
[72]	GANs	Network traffic anomaly detection	Not specified	Applies GAN-based models combining feature extraction and generation	No comparison to state-of-the-art DL; lacks adversarial defense integration
[21]	GANs	Survey on GANs for IDS	CICIDS-2017; NSL-KDD	Comprehensive survey; categorizes GAN types by threat model and dataset	Uses some outdated datasets; limited coverage of newer GAN variants or multimodal threats
[73]	GANs	IDS evaluation	NSL-KDD; CICIDS-2017	Evaluates WGAN, DCGAN, ACGAN with precision/recall	Dataset limitations; no longitudinal testing; GAN stability issues under-explored
[69]	GANs	IDS enhancement	CICIDS-2017	Proposes conditional GAN for attack pattern classification	No experimental implementation; minimal GAN configuration details
[74]	GANs	Phishing email defense	None	Generates phishing emails to harden classifiers via synthetic data	Email set lacks diversity; no comparison to transformer baselines
[75]	Adaptive GAN	Real-time threat detection	None	Adaptive GAN for evolving threats with minimal retraining	Not validated in dynamic/distributed environments; assumes feature stability
[76]	GANs, VAEs	Threat simulation (malware, phishing, intrusions)	Synthetic datasets	PCA-GAN hybrid for anomaly synthesis to train robust IDS	No real-time validation; limited dataset scope; no adversarial robustness checks
[67]	GANs, transformers	Anomaly detection	CICIDS-2017	Compares GANs vs transformers on intrusion data (detection rate, convergence, scalability)	Missing transformer tuning and GAN loss curves; reproducibility unclear
[68]	GANs, Chat-GPT	Cyber threat analysis	None	Framework outlining Gen AI misuse scenarios and attack simulation potential	No implementation or experiments

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Table A.3: Summary of state-of-the-art studies on Gen AI for cybersecurity.

Ref.	Gen AI alg.	Use case / domain	Dataset	Contribution	Limitations / challenges
[77]	GANs, GPT	GAN vs GPT in cybersecurity	CICIDS-2017; synthetic	Contrasts GAN and GPT for anomaly detection; reports accuracy/F1	Narrow dataset scope; simple GPT models; limited external validity
[78]	LLMs	Prompt injection, jail-breaking	None	Details jailbreak techniques and risks of unrestricted LLM use	No quantitative risk model; lacks detection/mitigation methods
[71]	LLMs	Defense simulation	None	Simulation architecture for cyber training using LLM outputs	Not evaluated; no red/blue team testing
[70]	LLMs, VAEs	Adaptive threat generation	None	LLM-driven adaptive response; feedback-based defense orchestration	No SIEM/SOAR integration; latency/performance metrics absent
[79]	ChatGPT	Explainable AI in cybersecurity	None	XAI-enhanced IDS with ChatGPT explanations; user-facing interpretability	Explanations not compared to ground truth/human baseline
[80]	ChatGPT	Social / ethical risks	None	Taxonomy of Gen AI misuse and threat classes; policy/ethics overview	Lacks empirical risk evaluation; conceptual focus
[81]	ChatGPT	Explainable IDS	CICIDS-2017	Uses ChatGPT to explain IDS decisions with LLM-based rationales	No adversarial evaluation; runtime overhead not reported; reliance on closed APIs
[82]	ChatGPT, LLMs	Policy and regulation risk	None	Analysis of Gen AI risks; layered defense and audit-based mitigation framework	No implementation/validation
[10]	ChatGPT, LLMs	Automated hacking / defense	None	Conceptual analysis: AI-generated phishing, malware, social engineering	No threat modelling or mitigation strategies
[83]	ChatGPT, Bard	LLM capabilities for offence/defense	None	Compares ChatGPT vs Bard for attack vector generation (evasiveness, diversity)	Dataset unspecified; metrics mostly qualitative

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